

Float reaction between sodium aluminoborosilicate glassmelts and copper-base liquid metal alloys

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Abstract

Wavelength-dispersive X-ray spectroscopy (WDS) has been utilized in a study of redox chemical diffusion dynamics in reaction couples of two sodium aluminoborosilicate glassmelts – both lightly doped with Fe^{3+} – with two copper-base liquid metal alloys: Cu–37 at.%Ge and Cu–36 at.%Sn. The glassmelts were floated on the alloys for 30 min in a controlled, reducing atmosphere at temperatures in the range 1200–1350 °C. Despite there being a significant driving potential to incorporate Cu^{+2+} into the glassmelt, such does not occur: ionic Cu concentration in the glassmelts was below the detection limits of WDS (i.e., $X_{\text{Cu}} < 100$ ppm atomic). As a generalization, the $\text{Ge}^{2+,4+}$ and $\text{Sn}^{2+,4+}$ diffused into the glassmelt to a depth of ~ 50 μm , a distance similar to that seen for ionic Sn in the soda-lime silicate on pure Sn float process that is operated at notably lower temperatures and for dramatically shorter reaction times. For reactions with Cu–37 at.%Ge, doping the glass with ferric iron had a profound effect on the depth of penetration of ionic Ge, decreasing it significantly: Fe^{3+} acts as an electron acceptor from Ge^{2+} ; the resultant tetravalent cation (Ge^{4+} , a network former) is all-but immobile. A similar result was not seen for reactions with the Cu–Sn alloy, a consequence of considerable reduction of the glassmelt by its pre-reaction with molybdenum components of a delivery system. The reaction morphologies seen are consistent with the Wagner/Schmalzried formulation for dynamic redox reactions.

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1. Introduction and scientific background

In the interests of preparing refractory, multicomponent flat glass for such high-value applications as flat-panel displays, we have pursued studies of the physical chemistry and reaction kinetics of the float-glass process. Initially developed and patented by Pilkington Brothers, Ltd. [1,2], the process traditionally involves floating a continuous ribbon of glassmelt across a bath of molten, high-purity tin in a controlled temperature

gradient such that the glassmelt stiffens as it progresses. The liquid metal thus transfers its flat surface to the glassmelt, creating smooth glass without the need for additional processing. To protect the molten metal, the float bath exists in a highly reducing environment. Technically, tin was chosen for the float medium based certain physicochemical properties that include (1) its low melting temperature (particularly as compared to the softening-point temperature of soda-lime silicate – NCS – glassmelts), (2) the large temperature range between its melting and boiling points, (3) its low vapor pressure at the range of temperatures associated with forming NCS glass (a corollary to point 2) and (4) the great differences in the standard Gibbs free energy of formation between the oxides of tin and those of the

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oxides that constitute NCS glass (e.g., $|\Delta_f G_{\text{SnO}}^0| \ll |\Delta_f G_{\text{SiO}_2}^0|, |\Delta_f G_{\text{Na}_2\text{O}}^0|, |\Delta_f G_{\text{CaO}}^0|$). Of course, economically, tin was chosen because it is inexpensive.

Regardless of point (4) above, the molten tin reacts with the glassmelt ribbon: ionic components of the glassmelt are reduced and added to the metal (making the float medium a dilute alloy); Sn cations are added to the glassmelt. Tin ions diffuse into the ribbon; spatially, the diffusion-limited reaction occurs both at the molten-metal/glassmelt contact surface (bottom) as well as at the top, free surface, the latter being a result of the significant vapor pressure of tin in the float bath. The extent of reaction is exacerbated with increased temperature: both the driving potential for the glassmelt/liquid-metal redox reaction as well as the mobility of ions in the glassmelt increase with temperature; in addition, a temperature-induced increase in the vapor pressure of tin impacts negatively – via reaction – the structural and thermal components of the float tank. Moving the temperature of the float bath to ever higher temperatures such that glasses more refractory than NCS can be so processed requires both an understanding of (1) the metal–silicate reaction thermodynamics and (2) the ionic (diffusion) dynamics in the silicate melt.

The reaction thermodynamics are easily understood by evaluating the impact of lowered activity on the oxidation/reduction reactions. For example, in the case of the standard NCS on pure Sn float reaction, the activity of metallic sodium in the liquid metal (a_{Na}) is diminishingly small; so, too, is the activity of stannous oxide in the NCS glassmelt (a_{SnO}). As a consequence, there exists an extremely large potential for a displacement reaction in which Na_2O is reduced to metallic Na, which alloys the Sn, and the oxygen so released reacts with the Sn such that SnO is added to the glassmelt. Similarly, a large potential exists for CaO and SiO_2 in the glassmelt to react as well, so as to increase the activities of Si and Ca in the float medium. (A wonderfully lucid description of the activity effect on redox-reaction thermodynamics can be found in the text by Paul [3], pp. 224–227.)

The kinetics of the float reaction are complicated. Analyses of the chemical profile-with-depth of NCS glass floated on Sn (i.e., in the commercial process) by a plethora of analytical techniques, including X-ray-emission and Mössbauer spectroscopies, reveal concentration profiles for both ionic Sn and Fe (the latter a minor chemical component of the glassmelt whose addition is related to fining) that are distinctly at odds with simple diffusion analyses based on Fick's 2nd law (see Ref. [4] for a synopsis). Specifically, concentration peaks for these elements are seen superposed on the Fick's-2nd-law profile: the peak for ionic tin occurs at 1/2 to 1/3 of its depth of total penetration; a peak for ionic iron occurs at the end-of-extent of tin penetration [5]. (Ionic tin penetration depths in commercial NCS float glass are

typically in the range 10–40 μm , the depth depending on the residence time of the glassmelt ribbon on the float bath.) Tin ions are primarily stannous (Sn^{2+}) near the free surface with stannic (Sn^{4+}) ions becoming more prevalent with depth; the internal $\text{Sn}^{2+,4+}$ peak corresponds to the maximum concentration of Sn^{4+} [6]; this peak also corresponds to a noted reduction of Fe^{3+} to Fe^{2+} [6,7].

Our group has explored both the thermodynamics and kinetics of the float reaction by pursuing experiments in which sodium aluminoborosilicate glassmelts (commercial Pyrex® [Corning Code 7740] and an Fe^{3+} -doped Code 7740 analog) were float-reacted on binary liquid metal alloys [4,8,9]. The glassmelt compositions were selected in order to work with compositions more refractory than NCS (NCS has a viscosity of 10^3 Pa s – the glassmelt viscosity employed at the hot, 'forming' end of the float bath – at 1100 °C; Code 7740 and the Fe^{3+} -doped 7740 analog have a 10^3 Pa s viscosity at 1255 °C and 1350 °C, respectively). The alloys used in our earlier work were gold-based alloys, Au–30 at.%Sn and Au–28 at.%Ge. Clearly, these alloys were not selected because they were potentially inexpensive(!); rather, the alloys were selected based on two scientific criteria: (1) these liquid alloys (solutions) are distinctly and strongly exothermic, meaning that the activities of Au and of Sn or Ge are significantly lower than their mole fraction (X_i) in solution (e.g., in Au–30 at.%Sn [$X_{\text{Sn}} = 0.30$] at 1300 °C, $a_{\text{Sn}} = 0.039$) and (2) given that the Au is essentially inert with respect to silicate melts [10], only one of the components of the alloy could react with the glassmelt. Lowering – and controlling by alloy chemical design – the activity of the reactive metal species diminishes the driving potential for reaction between the oxide and metal liquids; thus, exothermic alloying allows the temperature of the float reaction to be pushed to significantly higher temperature while not exacerbating the spatial extent of reaction. Code 7740 reacted with liquid Au–30 at.%Sn for 30 min at 1350 °C, for example, showed no more $\text{Sn}^{2+,4+}$ incorporation or penetration than NCS-on-Sn commercial float glass reacted for <1 min at 1100 °C. Choosing the reactive alloying element for the structural role its ionic form will play in the reacted glassmelt affects the extent of reaction as well: in the Au–28 at.%Ge experiments, Ge^{4+} acts as a network former in the glassmelt; oxidizing the ionic Ge to this state (a reaction controlled via Fe^{3+} additions to the glassmelt) makes it essentially immobile, to first order.

The results of the alloy experiments, combined with the published data of chemical profiles for commercial NCS-on-pure-Sn float glass, confirmed an application of Carl Wagner's theory for the dynamics of alloy oxidation [11], and an extension of this theory to oxidation and reduction reactions in crystalline ionic solids by Schmalzried [12,13], to describe holistically the kinetics

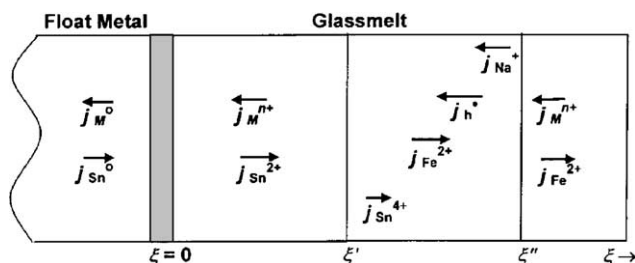


Fig. 1. Schematic diagram of the chemical/structural reaction fronts and associated chemical diffusion fluxes active in the cation-diffusion-limited reduction mechanism for an (Fe, Na)-bearing glassmelt floating on a Sn-bearing liquid metal. One surface and two internal reactions are indicated, at $\xi = 0$, ξ' and ξ'' ; arrows denote the direction of the flux for each species (after [4]).

of the float-glass reaction. The float-glass reaction is essentially understood as a dynamic reduction of the glassmelt [14] combined with solution formation, i.e., homogenization of the new added cations originally from the liquid-metal float bath. The most rapid dissipation of the reduction potential in the glassmelt is effected by the diffusive motion of the network-modifying cations (thus, cation diffusion is rate-limiting), motion that is charge-compensated both by the motion of electron holes ($h^\bullet$)¹ and by counter-motion of other cations. Fig. 1 presents the model schematically, for the case where ionic Sn is solely incorporated into the glassmelt by the reaction (e.g., for the commercial float reaction or for a reaction with a gold–tin alloy) [4]. The model involves multiple redox reactions occurring at the metal-melt/glassmelt interface and at two internal fronts; chemical diffusion between the external and internal interfaces ‘feed’ the continuing reaction overall. The internal interfaces are sites of chemical/structural reactions in the glassmelt; the non-Fick ‘humps’ in the chemical profiles (described above) can *only* be produced by such reactions. In words, the reaction at $\xi = 0$ is the redox exchange reaction described qualitatively above (in commercial NCS float, this reaction involves primarily the Na_2O component of the glassmelt: Na is so added to the Sn float bath) that incorporates Sn^{2+} into the glassmelt. The chemical diffusion between $\xi = 0$ and ξ' is by cation interdiffusion, while that between ξ' and ξ'' involves cation diffusion primarily charge-compensated by an ‘outward’ flux of $h^\bullet$. The internal reaction at ξ' consumes $h^\bullet$ in the transformation of Sn^{2+} to Sn^{4+} ; it is the loss of the electronic

defects at ξ' that forces the cation interdiffusion dynamics between $\xi = 0$ and ξ' . The reaction at ξ'' creates the mobile $h^\bullet$ and, in the process, ‘releases’ charge-compensating cations (primarily Na^+) that had served to stabilize Fe^{3+} in the network-former structural role. The ‘layered’ glass structure, implied clearly by the chemical data and described by this model, is neither static nor even steady-state; rather, it is a ‘dissipative’ [17], transient structure, frozen into the reacting system by its thermal history. Nevertheless, the model, and the philosophy supporting it, allows prediction and understanding, not only of the float-process proper, but of other, post-float chemical reactions as well (e.g., ‘bloom’) [18]. A float-reaction model with many similarities to ours was postulated independently by Heide, Frischat and co-workers [19,20].

This work reports on the next-level complication in the study of the float-glass reaction employing a heavily alloyed metal float medium: reactions in which both components of the binary alloy have the potential to react with the glassmelt. We selected two copper-base alloys, Cu–37 at.%Ge and Cu–36 at.%Sn; these were reacted with the same two aluminoborosilicate glassmelts used in the earlier studies. Both metal alloys are exothermic solutions, and both have a sufficiently low liquidus temperature to be of interest in float processing. Unlike the situation for gold, however, $\Delta_f G_{\text{Cu}_2\text{O}}^0 < 0$ for all temperatures examined experimentally; at the same time, $|\Delta_f G_{\text{Cu}_2\text{O}}^0|$ is notably smaller than $|\Delta_f G_{\text{GeO}}^0|$ and $|\Delta_f G_{\text{SnO}}^0|$ (i.e., Cu is more noble than are either Ge or Sn). Clearly, there is an economic aspect in the selection of Cu as an alloying element over Au.² Technically, however, working with multicomponent alloys in which a thermodynamic and kinetic ‘competition’ occurs with regard to how much of various alloy components get ionized and added to the glassmelt suggests a possibility of controlled ionic doping of the glass during float processing.

2. Experimental approach

2.1. Float alloys and glassmelt compositions

2.1.1. Cu–37 at.%Ge

In the Cu–Ge binary system, 37 at.%Ge is the eutectic composition in which a homogeneous liquid solidifies, at 644 °C, into an intermetallic compound ε_2 (approximate

¹ Iron-oxide-bearing aluminosilicate melts and glasses have limited electrical conductivity (from an engineering perspective they would be considered an insulator), but what conduction does occur is dominated by electron holes (small polarons). The $h^\bullet$ response has been characterized directly via Hall-effect measurements (e.g., [15]) as well as indirectly (but just as decidedly) via oxidation dynamics experiments [16]. The $h^\bullet$ dominance in the diffusion behavior was verified for $\text{Fe}^{2+,3+}$ concentrations as low as 0.04 at.%.

² For completeness sake, however, it should be noted that because the Au is insoluble in the glassmelt, no Au is ‘lost’ in the float reaction on the gold-base alloys. Further, the Au employed in the processing is fully (and easily, via oxidation of the liquid alloy) recoverable. Economically, then, the use of a gold-based alloy in float processing is a capital expense and not an operating one.

composition Cu_5Ge_2) plus virtually pure Ge. Cu–Ge liquid solutions have been studied in detail by a number of authors [21–24]. For use in this study, the data presented in these papers needed to be modeled and extrapolated. We treated the data as a temperature-sensitive sub-regular solution, treating the enthalpy of solution ($\Delta_{\text{mix}}H$) and the excess entropy of solution ($\Delta_{\text{mix}}S^E$) as temperature independent but compositionally sensitive – consistent with the published data. Third-order, Redlich–Kister-type polynomial fits to the data provided values for the solution parameters; the sub-regular model could then be extrapolated to any composition and temperature [18]. Fig. 2(a) presents the activity vs. composition diagram for liquid Cu–Ge alloys so calculated at 1300 °C while Fig. 2(b) illustrates the vapor pressure vs. temperature behavior for the Cu–37 at.%Ge solution in comparison with that for either pure end-member li-

quid. One sees distinct negative deviation from ideal solution behavior, characteristic of an exothermic solution, which manifests itself in a distinct lowering of vapor pressure.

Preparation of Cu–Ge alloy employed OHFC-grade Cu (in both wire and ~2-mm shot form) and ultra-high-purity Ge powder (<1 mm). The components were melted and homogenized in an alumina crucible for 1 h at 1300 °C and a partial pressure of oxygen (p_{O_2}) of $\sim 10^{-10}$ atm (maintained by a dynamic CO:CO₂ gaseous buffer).

2.1.2. Cu–36 at.%Sn

The binary phase diagram of the Cu–Sn (bronze) system includes numerous compounds, but includes, too, a eutectic reaction between the compound η ($\sim \text{Cu}_6\text{Sn}_5$) and pure Sn – a phase-diagram morphology indicative of an exothermic liquid solution. We selected the composition Cu–36 at.%Sn based on its liquidus temperature, ~ 675 °C; the composition thus insures that the alloy is still liquid at a temperature corresponding to a Code 7740 glassmelt viscosity of 10^{10} Pa s (which is the glassmelt viscosity corresponding to removal of the conventional NCS ribbon from the Sn float bath). The thermodynamic study of the Cu–Sn system guiding our calculations was performed by Shim et al. [25]. These data were again fit to third-order Redlich–Kister polynomials to perform temperature and composition extrapolation. Fig. 3 presents for this system and specific alloy the thermodynamic data paralleling that shown in Fig. 2 for the Cu–Ge solution.

In preparing the bronze alloy, the Cu source was as noted earlier and the Sn source was 99.7 at.%-pure, ~2-mm shot. The alloy was melted and homogenized for 20 min at 1100 °C, $p_{\text{O}_2} \sim 10^{-15}$ atm.

2.1.3. Glassmelt compositions

The compositions of sodium aluminoborosilicate glassmelts employed in these experiments are presented in Table 1; they are identical to those used earlier by Cook and Cooper [4,8]. 7740 is commercial Pyrex® (labeled ‘PT’ in the earlier papers); chemical analysis of this 7740 material revealed a total $\text{Fe}^{2+,3+}$ content of 0.006 at.%, with 70% of the iron as ferric. The Fe-7740 material (labeled ‘ST’ in the earlier papers) was prepared from AR-grade (or better) raw materials using a conventional melting approach; the total $\text{Fe}^{2+,3+}$ content of this iron-oxide-doped material was measured at 0.081 at.% with >80% of the iron as ferric. The addition to 7740 of the small amount of Fe_2O_3 made the glass slightly more refractory: the softening-point ($10^{6.6}$ Pa s) temperature of Fe-7740 is 840 °C, compared to ~ 820 °C for commercial 7740; the temperature difference at a 10^3 Pa s viscosity for the two compositions was cited earlier.

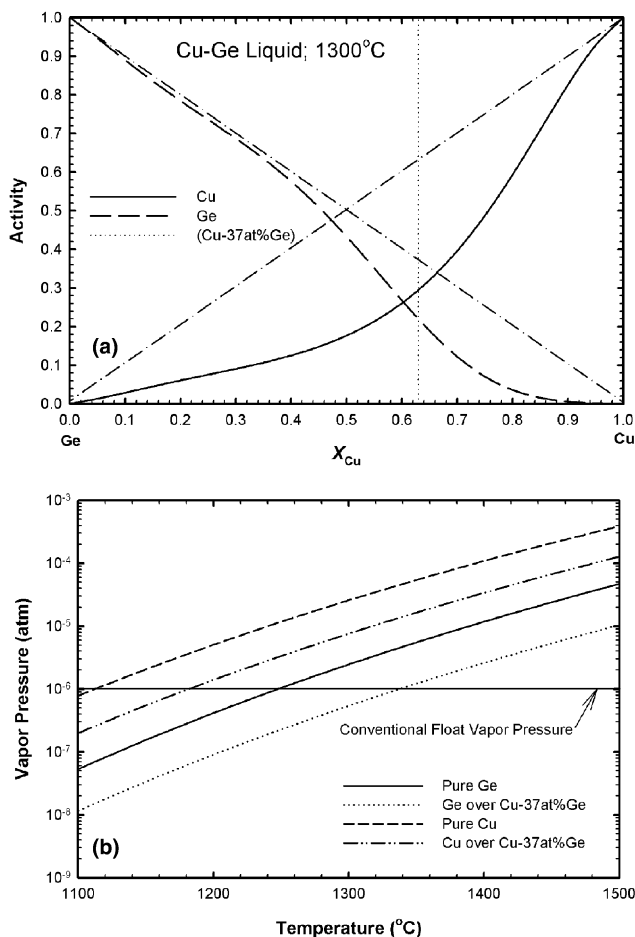


Fig. 2. Thermodynamic behavior of Cu–Ge liquid solutions. (a) Activity vs. composition (mole fraction) for Cu–Ge liquids at 1300 °C. For the alloy Cu–37 at.%Ge, a_{Cu} and a_{Ge} are 0.294 and 0.219, respectively. (b) Vapor pressures of components both in pure form and over the Cu–37 at.%Ge solution. The values are referenced to that associated with conventional float processing, i.e., for pure liquid Sn at 1100 °C, the saturated vapor pressure is 10^{-6} atm.

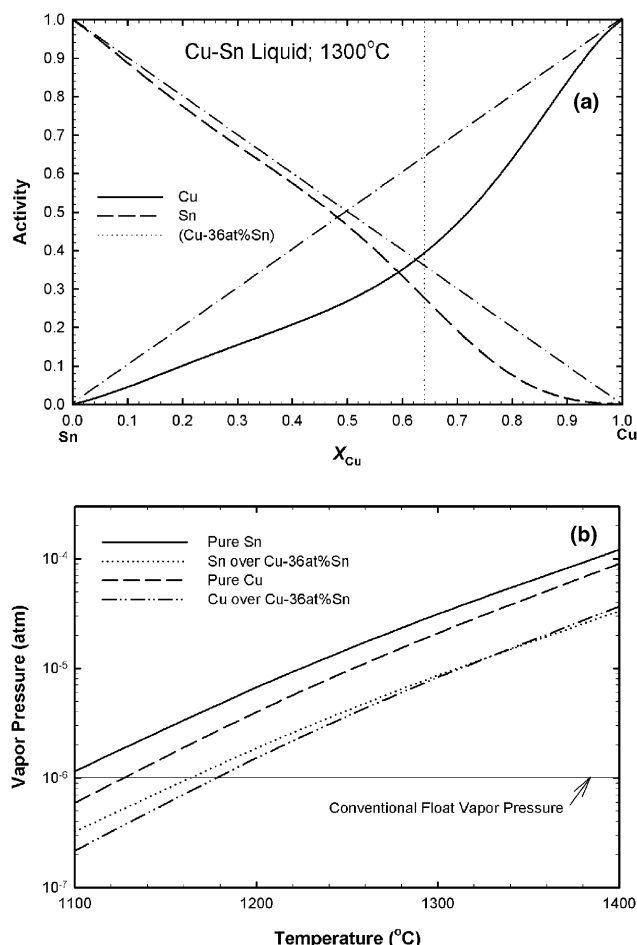


Fig. 3. Thermodynamic behavior of Cu–Sn liquid solutions. (a) Activity vs. composition (mole fraction) for Cu–Sn liquids at 1300 °C. For the alloy Cu–36 at.%Sn, a_{Cu} and a_{Sn} are 0.394 and 0.276, respectively. (b) Vapor pressures of components both in pure form and over the Cu–36 at.%Sn solution.

2.2. Apparatus and experimental protocol

The apparatus used in the experiments must allow a glassmelt to be reacted and cooled while in contact with an otherwise oxide-free liquid metal surface; our approach has been described in detail in Ref. [4]. In brief, the apparatus consists of a vertical tube furnace that incorporates a refractory cradle on a suspension elevator. Employing low-resistance o-ring seals and external linear bearings, the cradle can be translated vertically both rapidly and smoothly. The cradle, instrumented with a thermocouple, supports a 25-mm inner-diameter alumina crucible that serves as the float tank; $\sim 2 \text{ cm}^3$ of liquid metal alloy is employed in the experiments. The oxygen activity (a_{O_2}) in the apparatus is controlled with a flowing ($\sim 80 \text{ cm}^3 \text{ min}^{-1}$) $\text{CO}:\text{CO}_2$ buffer mixed at 10:1. The temperatures of glassmelt/molten-metal reactions studied here ranged from 1200–1350 °C, for which the dynamic gas buffer produces corresponding a_{O_2} 's of $\sim 1 \times 10^{-13}$ (1200 °C) to $\sim 8 \times 10^{-12}$ (1350 °C). These are

a_{O_2} values far below those at which the Cu, Ge or Sn components of the molten metal alloys would oxidize (e.g., for Cu–37 at.%Ge at 1200 °C the Ge component would not oxidize until the a_{O_2} exceeded $\sim 3 \times 10^{-10}$; similarly, for Cu–36 at.%Sn at 1200 °C, the Sn component would not oxidize until the a_{O_2} exceeded $\sim 1 \times 10^{-9}$).

Experiments reacting the glassmelts with the Cu–37 at.%Ge alloy followed the approach of the earlier studies. Specifically, the fire-polished, cylindrical glass specimens were first fused into a mullite ring (22-mm O.D.; 16-mm I.D.; 5-mm deep); the ring prohibits the glassmelt specimen from fusing to the alumina float vessel. At the beginning of the procedure, the ring-bound specimen rests on the lip of the float vessel, supported there by a 1.6-mm alumina rod. The suspension elevator is arranged such that the specimen/float-vessel assembly is approximately 13 cm above the hot zone of the furnace: with the hot zone set for the temperature of the reaction, the assembly experiences a temperature of ~ 800 °C at this location. The assembly is allowed to equilibrate at this temperature, and in the low- p_{O_2} environment created by the gas mixing, for 20 min, during which time the molten metal alloy is purged of all oxides (easily confirmed visually). The assembly is subsequently lowered into the hot zone (2 mm s^{-1}), the alumina support rod removed (with a tungsten finger) and the ring-contained specimen drops ($\sim 5 \text{ mm}$) onto the molten metal alloy. Glassmelt and molten metal were held in contact under these conditions for 30 min, after which, the assembly was lowered in 4-cm segments, producing rapid cooling in 200 °C steps, using $\sim 15 \text{ min}$ per step. Once the assembly temperature reached 700 °C, it was quickly lowered to the bottom of the assembly, the furnace opened and the specimen removed from the still-molten metal.

A gravity-fed delivery system – shown schematically in Fig. 4 – was developed for the experiments employing the Cu–36 at.%Sn alloy. The impetus for the innovation was the desire to alleviate the periodic trapping of gas bubbles at the glassmelt–metal melt interface in the ‘drop’ protocol described above; the design was successful in this regard. Implemented, the glassmelt is contained in a conical crucible, fabricated either from alumina or molybdenum, that incorporates a simple 6.4-mm diameter hole in its bottom. The data reported here were for reactions that employed molybdenum. The crucible is contained within a second refractory cradle, suspended above the float vessel. In that reaction/bonding of the delivered glassmelt to the wall of the float vessel remains undesirable, a mullite ring (of the same dimensions as described previously) is floated on the metal alloy melt to corral the delivered glassmelt. Two problems inherent in the design are (1) the inability to see the molten alloy as it is cleaned of oxide at 800 °C and (2) the loss of exactness in knowing when the sili-

Table 1
Experimental glassmelt compositions^a

Oxide	7740 ^b			Fe-7740 ^c		
	wt%	mol%	at.(ion)% ^d	wt%	mol%	at.(ion)% ^d
SiO ₂	81.14	83.45	25.66	82.31	84.72	26.09
B ₂ O ₃	12.70	11.28	6.94	12.40	11.02	6.79
Al ₂ O ₃	2.18	1.32	0.81	2.14	1.30	0.80
Na ₂ O	3.93	3.92	2.41	2.73	2.72	1.68
Fe ₂ O ₃	0.018	0.007	0.004	0.280	0.108	0.067
FeO	0.007	0.006	0.002	0.054	0.046	0.014
CaO	0.00	0.00	0.00	0.03	0.03	0.01
MgO	0.00	0.00	0.00	0.02	0.02	0.01
TiO ₂	0.01	0.01	0.00	0.03	0.02	0.01
O ²⁻			64.17			64.53
	%Fe as Fe ³⁺ = 70			%Fe as Fe ³⁺ = 82		

^a Major element wt% data from X-ray fluorescence (Corning, Inc., Corning, NY and XRAL Labs, Ltd., Hamilton, ONT); Fe²⁺:Fe³⁺ ratio data from wet-chemical titration (Corning, Inc.). Instrument stability and counting statistics errors on wt% data amount to $\leq 1\%$ (2σ) uncertainty. Mol% and at.% values are calculated from wt% data.

^b Commercially obtained Pyrex® rod.

^c Conventional melting of AR-grade (or better) oxides (NaCl as source of Na⁺) at 1600 °C for 16 h in air; annealed in air at 600 °C for 24 h.

^d For oxides, these values are the at.% for the cations (e.g., there is 25.66 at.% Si⁴⁺ in 7740); the at% of O²⁻ being listed in the last row.

cate-metal reaction begins. Consequently, reaction experiments were preceded by experimental calibration of the apparatus, discerning time and temperature sequences for the behavior of the float alloy and glassmelt relative to the two concerns. The 800 °C, 20-min treatment remained sufficient to clean the metal alloy; the 30-min reaction time of glassmelt with metal alloy at the reported experimental temperature we believe to be good to ± 5 min. (We have since improved the design of the delivery system such that a low-mass thermocouple in direct contact with the float vessel discerns rather precisely when the glassmelt comes into contact with the alloy.) The conclusion of the experiments with this apparatus follows exactly that of the ‘drop’ experiments.

Beyond visual inspection, the primary analytical tool employed in this study was electron probe microanalysis (EPMA; specifically wavelength-dispersive X-ray emission spectroscopy – WDS). EPMA specimens were prepared from cross-sectional cuts of the reacted glass samples; these sections were cold-mounted into a non-conducting polymer and polished to 0.05- μ m (levigated alumina). The specimens were subsequently carbon-coated (200–500 nm) to prevent charging under the electron beam. EPMA was performed on a Cameca SX-51 microprobe (five independent crystal spectrometers; 40° takeoff angle; acceleration voltage 15 kV; 0.5 μ m spot size creating an analysis volume of ~ 3 μ m diameter). K $_{\alpha}$ emission lines were employed for all cation analyses except for Ge, where the L $_{\alpha}$ line was used. Two different beam currents were utilized: 20 nA was used for Al, Fe, Cu, B and Ge analyses so as to provide accuracy at low photon counts; 1 nA was used for Si and Na analyses so as to minimize diffusive movement or

volatilization of Na⁺. Oxygen content was calculated by difference by assuming all Fe as divalent (FeO), all Ge and Sn as tetravalent (GeO₂ and SnO₂, respectively), all Mo as hexavalent (MoO₃) and all Cu as monovalent (Cu₂O). External standards employed included the unreacted 7740 and Fe-7740 glasses (for B, Al, Na and Si), AR-grade GeO₂ (for Ge), an Fe^{2+,3+}-bearing aluminosilicate glass (glass ‘HG’ from Ref. [14]; for Fe), natural cassiterite (SnO₂; for Sn) and natural cuprite (Cu₂O; for Cu).

3. Experimental results

3.1. Visual inspection and comment

Hand analysis of all reacted specimens revealed smooth, polished surfaces on both sides of the formed glass, i.e., the side that was the glassmelt-molten metal interface as well as the opposite, ‘free’ side. The metal-silicate reaction surface reflected the significant meniscus curvature of the molten metal alloy. Glassmelt specimens reacted with the Cu–37 at.%Ge alloy, following the ‘drop’ protocol, showed some fine bubbles (≤ 0.3 -mm diameter) on the float surface, concentrated near the mullite containment ring; some of these bubbles – immediately adjacent to the mullite – additionally penetrated the glassmelt to a height < 1 mm above the float surface. Such bubbles were not seen in experiments employing gravity-fed delivery of the glassmelt. For the 7740 glassmelt reacted with the Cu–37 at.%Ge alloy, there was also associated with those bubbles immediately adjacent to the mullite ring, a faint, reddish-orange discoloration. The discoloration was not seen for the

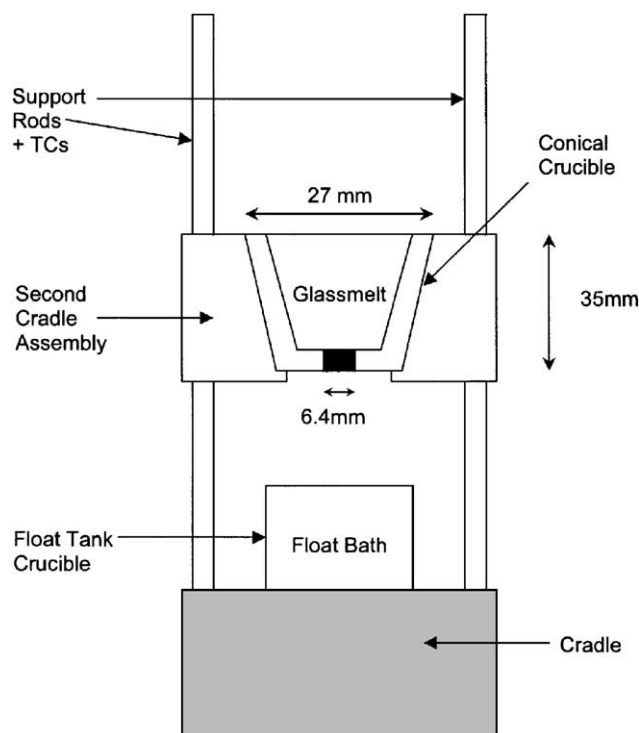


Fig. 4. Schematic diagram of gravity-fed delivery system. In this assembly, a mullite ring was placed within the float vessel to accept the delivered glassmelt. Molybdenum was used as the material for the conical crucible in the reaction experiments with Cu–36 at.%Sn reported here. An alumina crucible can be employed as well.

Fe-7740 glassmelt reacted with Cu–37 at.%Ge or for either glassmelt reacted with Cu–36 at.%Sn.

3.2. EPMA/WDS analyses

EPMA/WDS analyses of all specimens (Fe-doped or undoped glassmelt reacted with either of the copper-base alloys) found no copper ions in the reacted glassmelt. In that the resolution of WDS for Cu_2O is 0.02 wt%, this result means that the mole fraction of Cu^{+2+} in the reacted glassmelts does not exceed 10^{-4} (i.e., $X_{\text{Cu}} \leq 100$ ppm atomic). Additional care was taken to probe the regions of the 7740/Cu–37 at.%Ge reaction specimens where the discoloration was seen, with the exact same result with regard to Cu uptake of the glassmelt. We will comment on the distinctly limited Cu^{+2+} incorporation of the glassmelt below; nevertheless, this robust result allows a specific focus on the uptake of the alloying elements (i.e., Ge and Sn), via reaction, by the glassmelt.

Figs. 5 and 6 present representative EPMA/WDS chemical depth profiles for the glassmelts reacted with the Cu–37 at.%Ge and Cu–36 at.%Sn alloys, respectively. The standard (2σ) errors (based on X-ray counting statistics) for the $\text{Ge}^{2+,4+}$ and $\text{Sn}^{2+,4+}$ compositions shown in the figures are $\sim 5\%$, with the minimum detec-

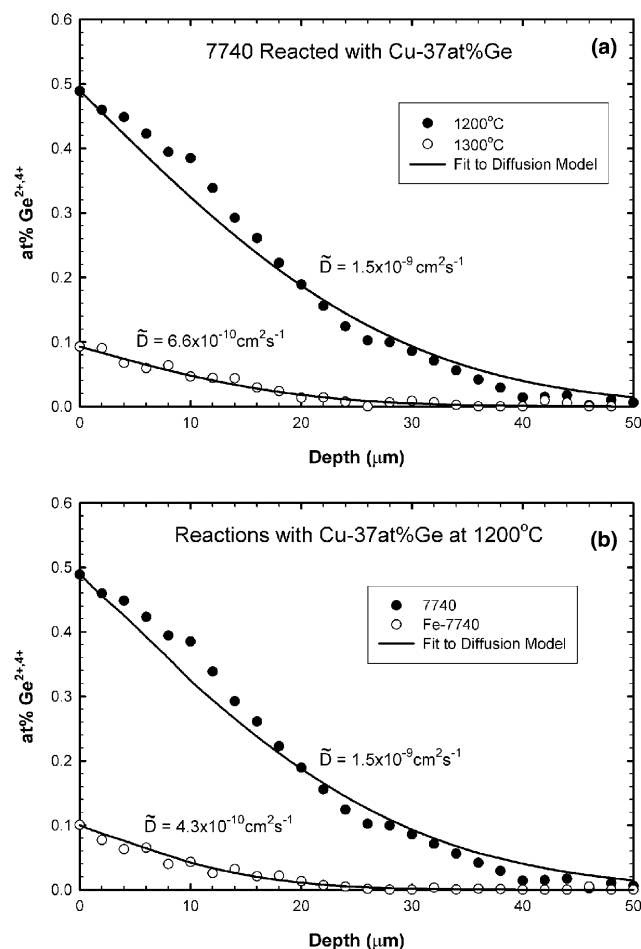


Fig. 5. EMPA/WDS profiles for $\text{Ge}^{2+,4+}$ diffused into 7740 and Fe-7740 glassmelts for 30-min reactions. Data are shown with non-linear least-squares fits to Fick's 2nd law. (a) Effect of temperature: an increase in temperature results in lowering significantly the solubility limit of ionic Ge in 7740; apparent diffusion coefficient is lowered as well. (b) Effect of Fe^{3+} doping: solubility and apparent diffusion coefficient of ionic Ge is lowered significantly. The poor quality of fit to the 7740/1200 °C data reflects the inadequacy of the simple Fick model to deal with internal redox phenomena that accompany the overall float-glass reaction.

tion limit of 0.017 at.% for both species [26]. These data are presented with non-linear, least-squares fits (Marquardt–Levenberg algorithm) to the Fick 2nd-law diffusion equation for a semi-infinite medium in contact with a source of fixed concentration:

$$C(\xi, t) = C_0 \operatorname{erfc}\left(\frac{\xi}{2\sqrt{\tilde{D}t}}\right), \quad (1)$$

where C is composition – with C_0 being the solubility limit in the glassmelt for the thermodynamic conditions employed in the experiment, ξ is the distance variable, t is time and $\tilde{D}$ is the chemical diffusion coefficient. In making the fits, C_0 and $\tilde{D}$ were employed as the adjustable parameters: the values of $\tilde{D}$ so calculated are noted explicitly in the figures, while the C_0 values can be discerned from the graphs.

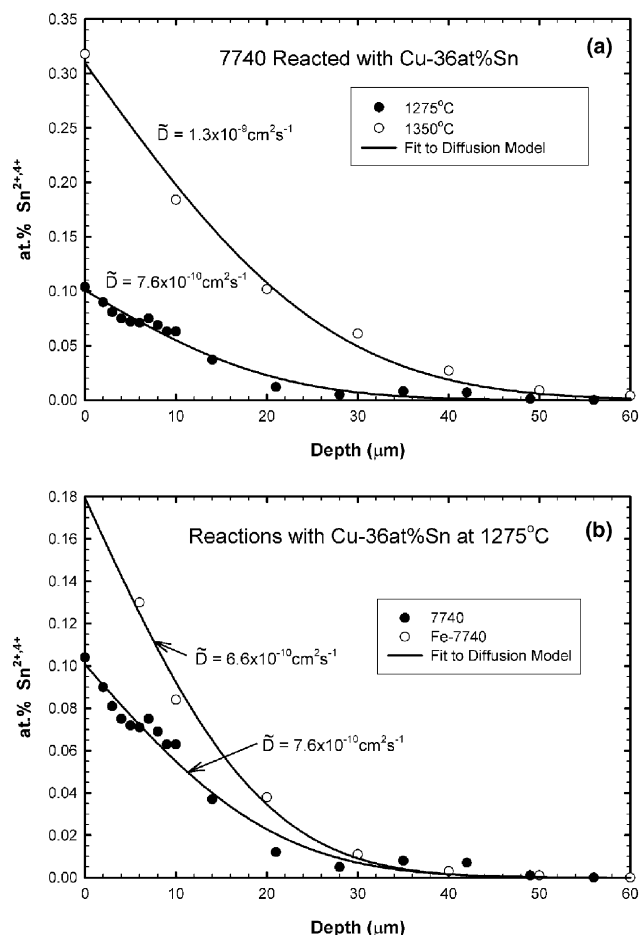


Fig. 6. EMPA/WDS profiles for $\text{Sn}^{2+,4+}$ diffused into 7740 and Fe-7740 glassmelts for 30-min reactions. Data are shown with non-linear least-squares fits to Fick's 2nd law. (a) Effect of temperature: an increase in temperature results in increasing the solubility limit of ionic Sn in 7740; the depth of ionic tin penetration seems affected but slightly. (b) Effect of iron-ion doping: the solubility of ionic Sn apparently increases, although depth of penetration is little affected. Both behaviors demonstrated here are different than in the case of the reactions with the germanium-bearing alloy; the inability of the addition of Fe^{3+} to the glassmelt to slow the reaction may be due to the (relatively) rapid reduction of the glassmelt via its reaction with the molybdenum crucible.

3.2.1. Glassmelt reactions with Cu-37 at.%Ge

For the glassmelts reacted with the Cu-37 at.%Ge alloy, Fig. 5(a) examines $\text{Ge}^{2+,4+}$ penetration in 7740 for 30-min reactions at 1200 °C and 1300 °C while Fig. 5(b) scrutinizes the difference in $\text{Ge}^{2+,4+}$ penetration between the 7740 and Fe-7740 glassmelts for a 1200 °C, 30-min reaction. Three facts concerning the reactions are manifest in the data: (1) an increase in temperature produces the surprising result that the extent of reaction – both the solubility limit for $\text{Ge}^{2+,4+}$ as well as the depth of penetration – is diminished; (2) the fit of the Fick model to the data – particularly well resolved for the reaction with 7740 at 1200 °C – shows a positive deviation of the data at shallow depth; and (3) at

constant temperature, an increase in Fe^{3+} in the glassmelt similarly diminishes the reaction.

3.2.2. Glassmelt reactions with Cu-36 at.%Sn

For the glassmelts reacted with the bronze (Cu-36 at.%Sn) alloy, the effect of temperature (Fig. 6(a)) on the reaction with 7740 seems 'better behaved' than those for the reactions with the Cu-Ge alloy: an increase in temperature here seems to cause an increase in $\text{Sn}^{2+,4+}$ solubility and reaction depth. The surprising aspect of these bronze-reaction data – the way these data are different than those of reaction studies of these glassmelts with Au-Sn, Au-Ge and Cu-Ge alloys – is in the comparison of the reactions with 7740 and Fe-7740 (Fig. 6(b)). In the previous investigations, doping the glassmelt with Fe^{3+} forced the divalent-to-tetravalent change of the cation incorporated from reaction with the float alloy to occur nearer to the glassmelt–float alloy interface, the results of which included a lower solubility limit for the incorporated cation and a distinct diminishment of its depth of penetration; such is not the case here (e.g., compare, qualitatively, Figs. 6(b) and 5(b)).

4. Discussion

4.1. Glassmelt reactions with Cu-37 at.%Ge

We evaluate the physical chemistry and kinetics of the reactions with Cu-37 at.%Ge by addressing directly the three enumerated points in Section 3.2.1.

We have witnessed previously – in experiments where these sodium aluminoborosilicate glassmelts were reacted with a molten Au-28 at.%Ge alloy [8] – the limiting of reaction extent with increasing temperature.³ This behavior is, in general, counter-intuitive: the transport coefficients for diffusive motion of all species in the glassmelt increase exponentially with temperature; further, incorporation of minor amounts of a 'new' ion into the glassmelt solution has such a strong, positive entropic effect that even a significantly exothermic solution enthalpy for the float alloy is overwhelmed. One is forced to conclude, then, that in the present experiments the reaction driving potential between the metal alloy float medium and the glassmelt diminishes significantly as temperature increases. Such behavior is possible if either the metal alloy solution becomes more exothermic with increasing temperature, or if the excess entropy of metal alloy solution increases (i.e., becomes less negative or, perhaps, more positive) with temperature. There is

³ The result is quite robust and replicable. The reaction(s) studied in these float experiments are extreme ones, that is, their driving potentials are very large. Consequently, the solution thermodynamics of both reactant phases are being probed – in a manner arguably described as 'exaggerated' – in these experiments.

some experimental evidence that liquid Au–Ge alloys might demonstrate increased exothermicity with increasing temperature [27] (indeed, this evidence was cited in the interpretation of our earlier data [8]), but this conclusion is based solely on comparing experimental results from different laboratories, and no similar evidence exists for Cu–Ge liquid alloys. An impact of excess entropy changes with increasing temperature is more likely, however.

Binary liquid alloys incorporating Group 14 elements, and particularly Si and Ge (and sometimes Sn), demonstrate complex structure. Si and Ge both undergo a semiconductor to metallic transition upon melting (e.g., [28,29]); both experience significant structural change as well – tetrahedral atomic coordination as crystals, coordination between 5 and 6 as a liquid. Despite the electronic behavioral change at melting, the limited atomic coordination means that bonds in the liquid remain significantly covalent; accordingly, reactions with ‘simple’ metals like Cu are highly exothermic. While thermodynamic experimental studies have suggested that liquid Cu–Ge alloys contain molecular clusters of approximate stoichiometry Cu_3Ge (e.g., [21]), recent X-ray diffraction analyses have confirmed the clusters’ presence [30]. Consequently, the mixing of three ‘species’ (Ge atoms, Cu atoms and the molecular clusters) in the binary alloy liquid allows for significant excess entropy of solution: at $\sim 1400^\circ\text{C}$, for example, $\Delta_{\text{mix}}S^{\text{E}}$ is positive between $0.30 \leq X_{\text{Ge}} \leq 1$, reaching a maximum of $1.17 \text{ J mol}^{-1} \text{ K}^{-1}$ (0.28 eu) at $X_{\text{Ge}} = 0.55$; between $0 \leq X_{\text{Ge}} \leq 0.3$, $\Delta_{\text{mix}}S^{\text{E}}$ is negative [21]. The behavior is logical: a composition near $X_{\text{Ge}} = 0.25$ should be mostly molecular clusters and overall solution entropy should be less than that associated with an ideal solution. Cluster formation, however, is a highly exothermic reaction: increasing the temperature thus breaks-up clusters and so forces the $\Delta_{\text{mix}}S^{\text{E}}$ to positive values. Thus, the temperature sensitivity of the float reaction studied here is logical, too: as temperature is increased, the positive $\Delta_{\text{mix}}S^{\text{E}}$ forces the $\Delta_{\text{mix}}G$ of the metal alloy to lower values, which is exactly the behavior seen experimentally [21]; in this way, reactivity of the liquid alloy with the glassmelt is lowered.

The other two aspects of the data were encountered previously in float-reaction studies with the Au-based alloys [4,8]. The positive deviation from the Fick model is consistent with the internal redox reaction, in this case the oxidation of Ge^{2+} to Ge^{4+} by the donation of electrons to ferric iron ions, i.e., similar to the reaction producing the tin cation hump in standard NCS float glass. Doping the glassmelt with excess ferric iron forces this reaction (which occurs at the internal interface noted in Fig. 1 as ξ') to occur closer to the glassmelt–float alloy interface. Given the distinctly lower transport coefficient for the network-forming Ge^{4+} , both the depth of penetration and the gross solubility are affected.

4.2. Glassmelt reactions with Cu–36 at.%Sn

While the effect of temperature on the reaction extent for the bronze alloy reactions (Fig. 6(a)) seems intuitive (i.e., greater reaction extent at temperature increases), the magnitude of the change requires consideration. From the values of effective $\bar{D}$ discerned from the Fick analysis ($7.6 \times 10^{-10} \text{ cm}^2 \text{ s}^{-1}$ at 1275°C and $1.3 \times 10^{-9} \text{ cm}^2 \text{ s}^{-1}$ at 1350°C), one can calculate an apparent activation energy for diffusion of $\sim 150 \text{ kJ mol}^{-1}$ for these data – a value that can be compared to activation energies for tracer diffusion of divalent network-modifier cations in a variety of similarly polymerized aluminosilicate melts, which are in the range 200 – 300 kJ mol^{-1} [31]. The lower apparent activation energy evident in the present data again indicates that the increase in temperature diminishes the driving potential for the reaction between float alloy and glassmelt, albeit to a lesser extent than in the case of reaction with the Cu–Ge alloy. Evaluation of the Cu–Sn liquid solution model proposed by Shim et al. [25] indicates a positive excess entropy ($\Delta_{\text{mix}}S^{\text{E}} = 1.11 \text{ J mol}^{-1} \text{ K}^{-1}$ for Cu–36 at.%Sn); the data suggest the presence of Cu_3Sn -like atomic clusters in the melt – paralleling the behavior of the Cu–Ge alloys described previously.

The apparent – and unique from the perspective of all our experiments of float-glass reactions with heavily alloyed metals – ineffectiveness of the original Fe^{3+} content of the Fe-7740 glassmelt to suppress the extent of reaction with the bronze alloy commands reflection. The fundamental difference between these experiments and all the previous ones is the use of molybdenum as the crucible in the delivery system. WDS of these glasses reveal incorporation of some $\text{Mo}^{2+,4+}$ ions in the near surface (in the range 1.8 at.% at $5 \mu\text{m}$ depth to 0.3 at.% at $30 \mu\text{m}$ depth in Fe-7740 reacted at 1275°C for 30 min); this addition almost certainly involved the reduction of Fe^{3+} to Fe^{2+} .⁴ The glassmelt, so reduced at a rate faster than the incorporation of tin cations, will see greater incorporation of Sn^{2+} via interdiffusion with Fe^{2+} (compare to the case of redox kinetics in most aluminosilicate melts, where the most efficient reaction mechanism is one rate-limited by chemical diffusion of the divalent network-modifying cations [32]). Indeed, recent float-reaction experiments (to be reported elsewhere) where glass–ceramic-forming, transition-metal-cation-doped magnesium aluminosilicate melts were delivered onto the bronze alloy via an alumina crucible have displayed reaction morphologies similar to that seen for the previous studies, i.e., a first-order limitation

⁴ The observation supporting this argument is the incompatibility of similar glassmelt compositions – but containing minor amounts of nickel cations (as colorant dopants) – with molybdenum melting electrodes: the $\text{Ni}^{2+,3+}$ reduce, turning the glassmelt black (R.R. Thomas, Corning, Inc., personal communication).

on the reaction extent through imposition of a ‘forced’ internal redox couple.

4.3. The incorporation of copper ions into the glassmelt

In order for the interface between glassmelt and copper-base liquid alloy to form a local equilibrium (a requirement of the continuing reaction), some copper must be oxidized and enter the glassmelt as Cu^+ (and, perhaps, be there oxidized to Cu^{2+}). We were, nevertheless, surprised at the very low solubility of copper ions in these aluminosilicate melts: such modest solubility suggests clearly that the activity coefficient for Cu_2O in the silicate solutions exceeds unity, most likely significantly. Direct thermodynamic studies of the solution behavior have confirmed the fact: for calcium–magnesium aluminosilicate melts, the activity coefficient for $\text{CuO}_{0.5}$ (i.e., for Cu^+ in a solution model emphasizing an accounting of the number of cations) was measured as 10 ± 1 at 1300°C [33–35]. Further, these studies demonstrate conclusively (1) the activity coefficient for $\text{CuO}_{0.5}$ diminishes as the activity of SiO_2 increases, which means that a higher-silica-activity glassmelt will incorporate more Cu^+ , and (2) an increase in oxygen activity of the glassmelt will increase, too, its solubility of Cu^+ . These two facts explain the presence of some red discoloration of the 7740 glassmelt when it was reacted with the Cu–Ge alloy – coloration located specifically where the glassmelt contacts the mullite ring – as well as the lack of similar coloration in the other experiments.

In the 7740/Cu–37 at.%Ge reactions, a relatively reduced glassmelt (i.e., initially lower Fe^{3+} concentration than in Fe-7740) is placed in contact with both the float alloy and, at the containment ring, with mullite. The mullite tube used to fabricate the rings (MV30; Vesuvius-McDaniel, Beaver Falls, PA) has a (simplified) chemistry of approximately 51 mol% SiO_2 and 49 mol% Al_2O_3 [36]; at the temperature of these experiments, the rings consist of two phases, the mullite solid solution plus a glass of essentially pure silica composition (the equilibrium assemblage would be silica-saturated mullite plus cristobalite). At the interface of glassmelt, mullite ring and float alloy, the activity of silica is unity. Thus, relative to far away from the mullite, the glassmelt will incorporate a slightly higher concentration of copper ions, consistent with the silica activity effect; the total amount of Cu^{+2+} remains at the limit of EPMA/WDS analysis, but the impact of this locally enhanced copper-ion incorporation on color becomes evident. The lack of near-mullite-ring coloration of the other reaction specimens must involve subtle effects of the glassmelt chemistry. For example, the lack of near-ring color in the Fe-7740/Cu–37 at.%Ge specimens may result from a limitation of the solubility of copper ions due to the near-surface oxidation of Cu^+ to Cu^{2+}

and the subsequent effect on ion mobility (i.e., similar to the effect on tin ions). Continuing, the lack of coloration in the glassmelts reacted with Cu–36 at.%Sn must be a result of the same significant reduction effect caused by contact of the glassmelt with the molybdenum delivery system, described earlier, and this despite the higher activity of copper in the Cu–Sn alloy compared to the Cu–Ge alloy. In a simple test of this molybdenum-reduction hypothesis, the 7740 glassmelt was float-reacted with the Cu–Sn alloy (a single reaction at 1350°C for 30 min), but an alumina crucible was employed in the delivery system: for these conditions, faint red coloration was seen emanating from the glassmelt/float alloy/mullite-ring contact [18].

5. Summary/conclusions

We have pursued reaction studies in which Pyrex™-composition glassmelts have been floated on exothermic Cu–Ge and Cu–Sn metal alloys. Isothermal float reactions with temperatures in the range 1200 – 1350°C and durations of 30-min display, in general, reaction depths similar to those of commercial soda-lime silicate (NCS) glassmelts floated on a pure Sn medium but reacted at temperatures no higher than 1100°C and for durations measured in seconds. The differences in reaction extent convolve the properties of the glassmelts – in this case more refractory, and thus having a lower ionic mobility, than NCS at a fixed temperature – with those of the float medium: the exothermic alloys are less reactive with the glassmelt than is pure tin. Despite an obvious driving potential for the incorporation of copper ions into the glassmelt, such incorporation is very limited: $X_{\text{Cu}} \leq 100$ ppm atomic. Subtleties of the reaction, as evidenced by some red coloration in some specimens but at a specific specimen location, indicate the role of silica activity $\pm$ oxygen activity on the incorporation of the very minor Cu_2O component. Overall, the experiments articulated here confirm the dynamic of the float-glass reaction consistent with the Wagner/Schmalzried formalism and articulated by Cook and Cooper [4]. In addition, these experiments illustrate a fruitful approach to the design of float alloys and glassmelts such that control of specific chemical activities allows chemical design, via reaction doping, of the float-glass surface. Application of the approach to floating/float-reacting (and thus doping) glass–ceramic-forming glassmelts is the emphasis of continuing research.

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